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Metody komputerowe w spalaniu

Numerical Study of Hydrogen-Enriched Methane–Air Combustion

Using Detailed Chemical Kinetics

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Abstract

This work investigates the effect of hydrogen enrichment on the combustion of methane–air mixtures using detailed chemical kinetics in Cantera with the GRI-Mech 3.0 mechanism. Hydrogen fraction in the fuel was varied from 0 % to 40 % (molar) across a range of equivalence ratios. Adiabatic equilibrium states, autoignition delay times, laminar flame speeds, and a perfectly stirred combustor were analysed. At $\phi = 1.0$, 300 K and 1 atm, the calculated laminar flame speed increased from 37.68 cm s^{-1} for methane–air to 52.16 cm s^{-1} for 40 % hydrogen in the fuel. At $T_0 = 1000 \text{ K}$, $p = 10 \text{ atm}$ and $\phi = 1.0$, the constant-pressure ignition delay decreased from 81.99 ms to 12.64 ms over the same enrichment range. The perfectly stirred reactor results identify stability-transition brackets between cold extinguished and hot stable steady-state solutions on the tested residence-time grid.

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Nomenclature

Symbol	Meaning
ϕ	equivalence ratio
τ_{ign}	ignition delay
τ_{res}	residence time
S_L	laminar flame speed
T_{ad}	adiabatic flame temperature
X_i	mole fraction of species i
Y_i	mass fraction of species i
p	pressure
T_0	initial temperature
$\dot{m}$	mass-flow rate

1 Introduction

Hydrogen is widely studied as a clean fuel for gas-fired combustors and turbines, and blending hydrogen into methane is a practical near-term route to lowering carbon emissions from existing equipment [1]. Published experimental and review studies report that hydrogen enrichment generally increases laminar burning velocity and changes ignition and flame stability characteristics. Its implications for flashback and nitrogen-oxide formation depend strongly on mixture composition and operating conditions [1, 2].

In an engineering context, hydrogen can be added to existing gaseous fuels as part of fuel-flexibility or decarbonisation studies for gas-turbine or aero-engine combustor concepts. Such changes affect ignition propensity, laminar flame speed, flame temperature and combustion stability. The models used here are therefore treated as preliminary trend-analysis tools for CH_4/H_2 -air mixtures, not as a complete combustor model.

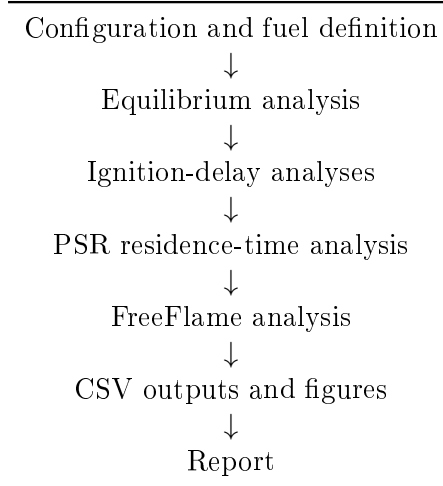
The objective of this study is to quantify, with a single consistent kinetic model, how the molar hydrogen fraction in a CH_4/H_2 fuel blend affects: (i) the adiabatic flame temperature and equilibrium product composition; (ii) the autoignition delay time and its sensitivity to temperature, pressure and equivalence ratio; (iii) the laminar flame speed; and (iv) the behaviour of an ideal perfectly stirred combustor on a prescribed residence-time grid.

2 Methodology

2.1 Project workflow

The calculations are organised around a common definition of fuel composition, oxidizer composition and operating-condition grids. Table 1 summarises the computational workflow; it is a project organisation diagram rather than a statement that all models are physically coupled in sequence.

Table 1: Computational workflow used in the project.



2.2 Chemical mechanism

All simulations were performed with Cantera [3] using the GRI-Mech 3.0 mechanism [4], an optimised detailed mechanism for natural-gas combustion comprising 53 species and 325 reactions and including nitrogen chemistry for NO formation. The mechanism is well suited to methane-dominated mixtures; its applicability to high hydrogen fractions is discussed in Section 4.1.

2.3 Fuel mixture definition

The fuel is a binary CH_4/H_2 blend parameterised by the molar hydrogen fraction

$$x_{\text{H}_2} = \frac{n_{\text{H}_2}}{n_{\text{H}_2} + n_{\text{CH}_4}}, \quad (1)$$

so that the fuel composition is $\{\text{CH}_4 : 1 - x_{\text{H}_2}, \text{H}_2 : x_{\text{H}_2}\}$. Air is modelled as $\text{O}_2 + 3.76 \text{N}_2$, and the mixture is set with a prescribed equivalence ratio ϕ .

2.4 Adiabatic equilibrium

Equilibrium states are obtained at constant enthalpy and pressure (HP problem), giving the adiabatic flame temperature and the equilibrium mole fractions of NO, CO, CO_2 and H_2O . The initial state is $T_0 = 900 \text{ K}$ and $p = 10 \text{ atm}$. The tested equivalence-ratio grid is $\phi = 0.6, 0.8, 1.0, 1.2$ and 1.4 , and the hydrogen fractions in the fuel blend are 0, 10, 20, 30 and 40 mol %.

2.5 Autoignition delay

Autoignition is computed in a homogeneous adiabatic reactor, modelled both as a constant-volume (`IdealGasReactor`) and a constant-pressure (`IdealGasConstPressureReactor`) system. The ignition delay τ is defined as the time of maximum temperature-rise rate, $\max(dT/dt)$; the

time of peak OH mass fraction is used as an independent cross-check. The primary temperature sweep uses the constant-pressure reactor at $\phi = 1.0$, $p = 10$ atm, $T_0 = 900, 950, 1000, 1050$ and 1100 K, with hydrogen fractions of 0, 10, 20, 30 and 40 mol %. The pressure sweep uses $T_0 = 1000$ K, $\phi = 1.0$, pressures of 5, 10 and 20 atm, and hydrogen fractions of 0, 20 and 40 mol %. The equivalence-ratio sweep uses $T_0 = 1000$ K, $p = 10$ atm, $\phi = 0.6$ to 1.4 in increments of 0.1 , and hydrogen fractions of 0, 20 and 40 mol %.

2.6 Laminar flame speed

The laminar flame speed is obtained from a one-dimensional freely propagating premixed flame (**FreeFlame**) with adaptive grid refinement. The final calculations use multicomponent transport with Soret diffusion enabled. The reported speed is the unburned-gas inlet velocity of the converged solution. Reference conditions are an unburned temperature of 300 K and a pressure of 1 atm, chosen so that the stoichiometric methane–air result can be used as a reference consistency check (Section 3.7). The domain width is 0.03 m; the tested equivalence ratios are $\phi = 0.6, 0.8, 1.0, 1.2$ and 1.4 ; and the hydrogen fractions are 0, 20 and 40 mol %. Grid refinement uses ratio = 3.0, slope = 0.06 and curve = 0.12.

2.7 Perfectly stirred reactor

A perfectly stirred reactor (PSR) is built from an inlet reservoir, an **IdealGasReactor** and an exhaust reservoir, coupled by a **MassFlowController** and a **PressureController**. The inlet mass flow is set as $\dot{m} = \frac{m_{\text{reactor}}}{\tau_{\text{res}}}$, allowing the residence time τ_{res} to be swept. The PSR is zero-dimensional, adiabatic and perfectly stirred. It is initialised from equilibrium products at the inlet enthalpy and pressure, then continued from long to short residence time; as a result, the calculated sequence follows the hot burning branch where that branch is available. The steady state is solved at each tested residence time to identify a stability-transition bracket between cold extinguished and hot stable steady-state solutions on the discrete grid. The inlet temperature is 1000 K, the pressure is 10 atm, the tested equivalence ratios are $\phi = 0.7$ and 1.0 , and the hydrogen fractions are 0, 20 and 40 mol %. The residence-time grid is 0.001 to 50 ms as configured, and the hot-gas residence time is $\tau = \frac{m_{\text{ss}}}{\dot{m}}$. Equilibrium-product initialisation is used before continuation. A 500 K temperature rise is used as the operational classification threshold between extinguished and stable cases.

2.8 Summary of analysis scope

Table 2 collects the operating ranges used in the individual calculations. The entries are taken from the project configuration and the generated CSV files; they also indicate which result from each model is used in the discussion.

Table 2: Scope of the numerical analyses.

Analysis	Model	Temperature	Pressure	ϕ	H ₂	Main result
Equilibrium	HP equilibrium	900 K	10 atm	0.6–1.4	0–40%	T_{ad} , NO, CO
Ignition	CP reactor	900–1100 K	10 atm	1.0	0–40%	ignition delay
CV/CP comparison	homogeneous reactors	900, 1000, 1100 K	10 atm	1.0	0, 20, 40%	CP–CV delay change
Pressure sweep	CP reactor	1000 K	5, 10, 20 atm	1.0	0, 20, 40%	pressure sensitivity
ϕ sweep	CP reactor	1000 K	10 atm	0.6–1.4	0, 20, 40%	delay and T_{max}
PSR	ideal stirred reactor	1000 K	10 atm	0.7, 1.0	0, 20, 40%	stability-transition bracket
FreeFlame	1-D premixed flame	300 K	1 atm	0.6–1.4	0, 20, 40%	laminar flame speed

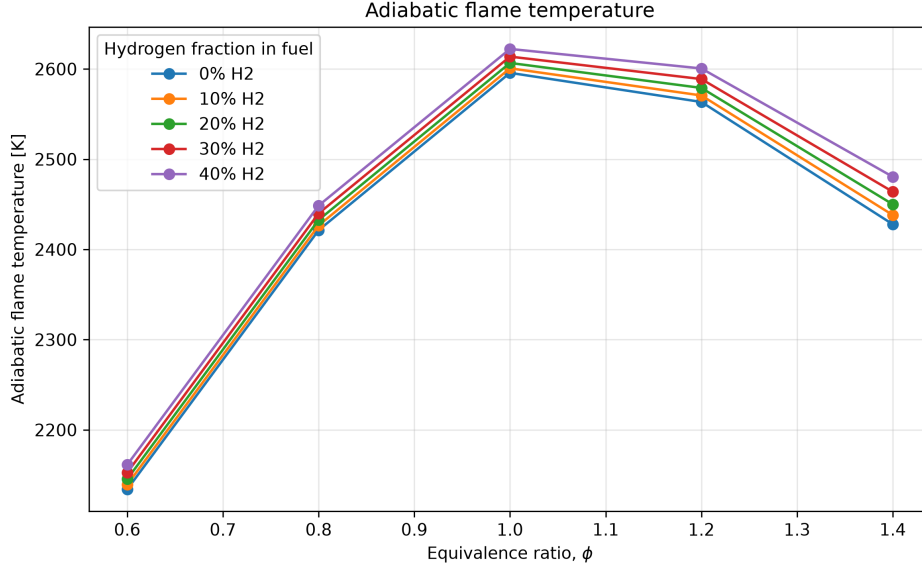


Figure 1: Adiabatic flame temperature versus equivalence ratio for different hydrogen fractions at $T_0 = 900$ K and $p = 10$ atm.

3 Results and discussion

3.1 Key numerical results

The main project-level results are summarised in Table 3. The PSR entry is intentionally reported only as a stability-transition bracket on the tested residence-time grid.

Table 3: Selected headline results for methane–air and the 40 % hydrogen fuel blend.

Quantity	Conditions	0% H ₂	40% H ₂
Laminar flame speed	$\phi = 1.0$, 300 K, 1 atm	37.68 cm s ⁻¹	52.16 cm s ⁻¹
Ignition delay	$\phi = 1.0$, 1000 K, 10 atm	81.99 ms	12.64 ms
Stoichiometric T_{ad}	HP equilibrium, 900 K, 10 atm	2595.74 K	2622.25 K
PSR stability-transition bracket	$\phi = 1.0$, 1000 K, 10 atm	0.002–0.005 ms	0.002–0.005 ms

3.2 Adiabatic flame temperature and equilibrium composition

On the tested five-point equivalence-ratio grid, the maximum adiabatic flame temperature is observed at $\phi = 1.0$ for every fuel blend (Figure 1). For methane–air, T_{ad} increases from 2134.18 K at $\phi = 0.6$ to 2595.74 K at $\phi = 1.0$, then decreases to 2427.90 K at $\phi = 1.4$. With 40 mol % hydrogen in the fuel, the corresponding values are 2161.47 K, 2622.25 K and 2480.57 K. Hydrogen enrichment therefore raises the stoichiometric equilibrium temperature by 26.51 K over the tested range.

The equilibrium NO mole fraction is largest near the lean side of the tested range: for methane–air it is 0.007842 at $\phi = 0.8$, 0.004817 at $\phi = 1.0$ and 0.000163 at $\phi = 1.4$. At 40 mol % hydrogen, the corresponding values are 0.008123, 0.004987 and 0.000224. Equilibrium CO is very small on the lean side and increases strongly for rich mixtures, from 0.000256 at $\phi = 0.6$ to 0.078503 at $\phi = 1.4$ for methane–air, and from 0.000266 to 0.067442 for the 40 mol % hydrogen blend. These equilibrium mole fractions must not be interpreted as actual engine emissions: they represent infinite-time adiabatic chemical equilibrium at fixed enthalpy and pressure, without finite residence time, turbulent mixing, heat loss, or exhaust quenching.

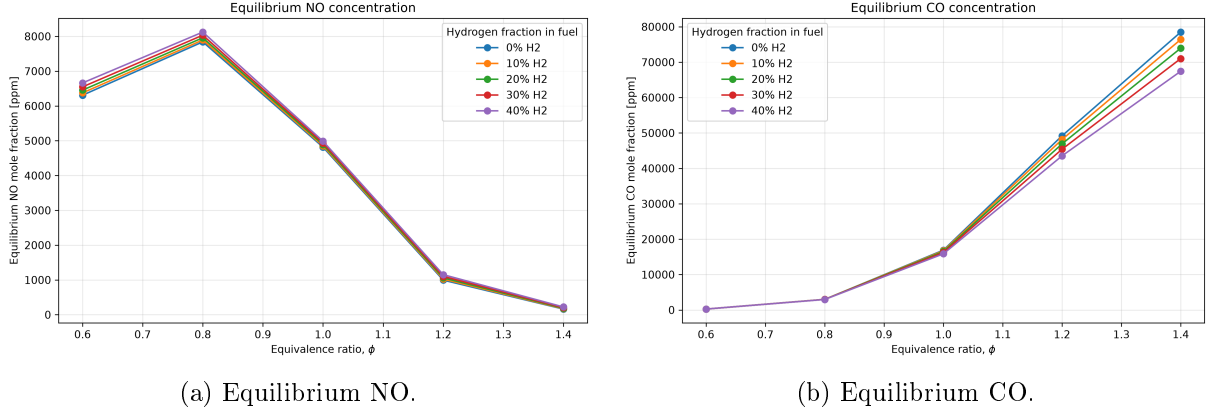


Figure 2: Equilibrium pollutant mole fractions versus equivalence ratio at $T_0 = 900$ K and $p = 10$ atm.

3.3 Nitrogen oxide formation

The equilibrium and PSR results report NO mole fractions from the GRI-Mech 3.0 chemistry. Established nitrogen-chemistry pathways in combustion are reviewed by Miller and Bowman [5]. The present study does not perform reaction-path, rate-of-production or sensitivity analysis, so it reports only the overall NO trends predicted by the selected mechanism. These values are not certified engine-emission indices.

3.4 Autoignition delay

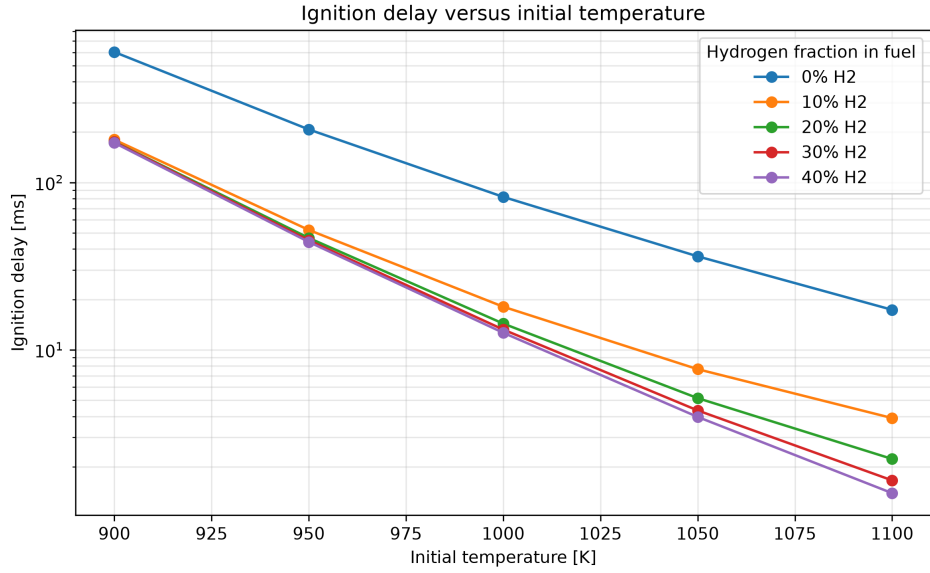
The ignition-delay results show a strong temperature dependence (Figure 3). At $\phi = 1.0$ and $p = 10$ atm, methane–air decreases from 601.98 ms at $T_0 = 900$ K to 17.34 ms at $T_0 = 1100$ K. The 40 mol % hydrogen blend decreases over the same temperature range from 172.58 ms to 1.39 ms. At $T_0 = 1000$ K, hydrogen enrichment reduces the delay from 81.99 ms at 0 mol % hydrogen to 18.09 ms at 10 mol %, 14.34 ms at 20 mol %, 13.21 ms at 30 mol % and 12.64 ms at 40 mol %.

At $T_0 = 1000$ K and $\phi = 1.0$, increasing pressure from 5 atm to 20 atm shortens the methane–air ignition delay from 178.55 ms to 39.61 ms. For 20 mol % hydrogen the same pressure increase reduces the delay from 18.85 ms to 9.98 ms, and for 40 mol % hydrogen from 14.62 ms to 9.24 ms.

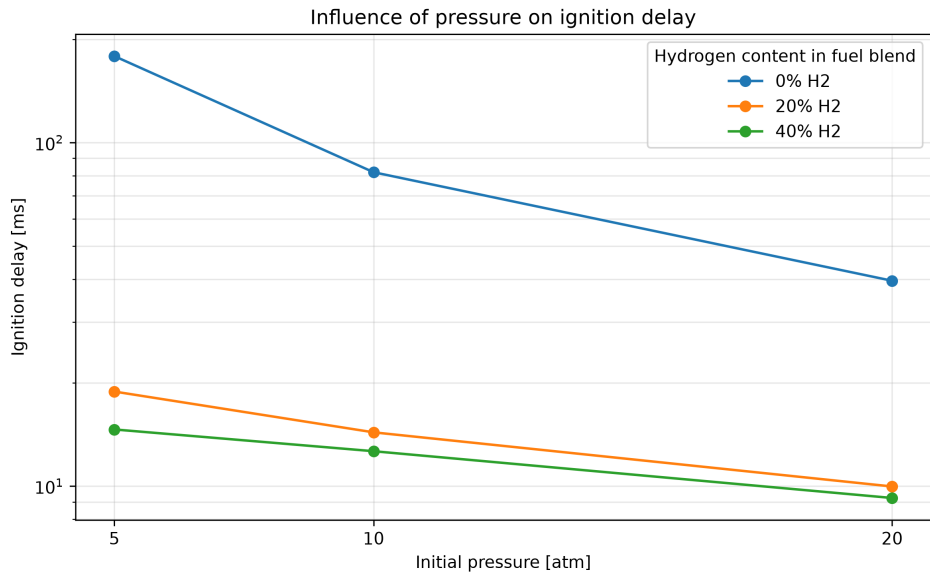
The equivalence-ratio sweep at $T_0 = 1000$ K and $p = 10$ atm shows different behaviour across fuel blends (Figure 5). Methane–air lengthens monotonically across the tested range, from 71.76 ms at $\phi = 0.6$ to 93.52 ms at $\phi = 1.4$. The hydrogen-enriched cases shorten as the mixture is enriched over the same range: the 20 mol % case changes from 18.41 ms to 12.37 ms, and the 40 mol % case from 16.36 ms to 10.76 ms. These trends are reported as outcomes of the present finite-rate simulations; detailed radical pathway attribution would require additional reaction-path or sensitivity analysis.

3.5 Two-dimensional ignition-delay map

A supplementary ignition-delay map was computed with the same homogeneous, adiabatic, constant-pressure reactor formulation as the one-dimensional ignition-delay sweeps. The pressure was fixed at $p = 10$ atm, the initial temperature was varied from 900 K to 1100 K in 25 K increments, and the equivalence ratio covered $\phi = 0.6$ –1.4 in increments of 0.1. The hydrogen fractions were 0, 20 and 40 mol % in the fuel. The map is intended for comparing finite-rate ignition trends across this idealised parameter space; it is not an operational map of a real



(a) Versus initial temperature at $\phi = 1.0$ and $p = 10$ atm.



(b) Versus pressure at $T_0 = 1000$ K and $\phi = 1.0$.

Figure 3: Autoignition delay time for the constant-pressure reactor.

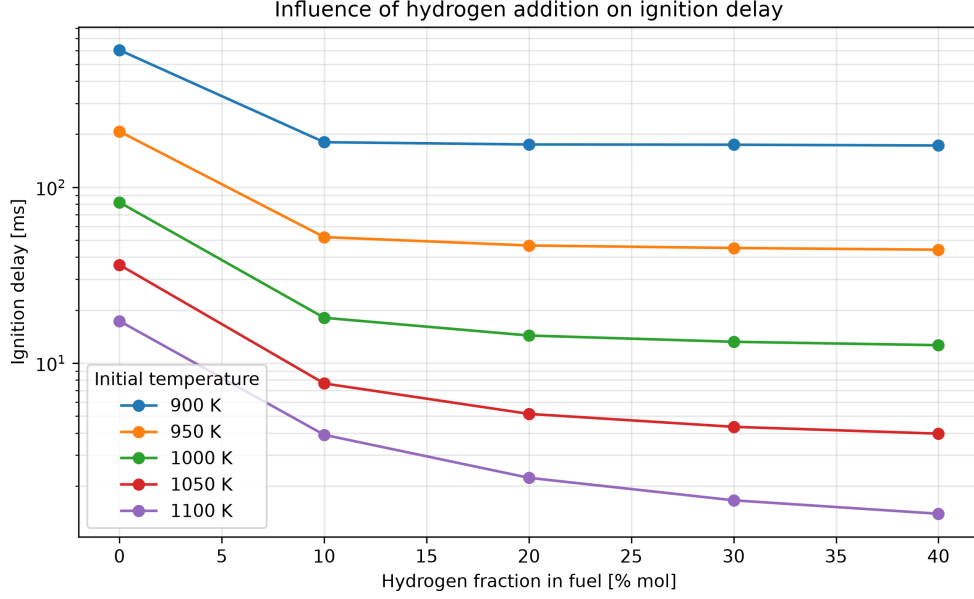


Figure 4: Constant-pressure ignition delay versus hydrogen fraction at $\phi = 1.0$, $p = 10$ atm and $T_0 = 900$ – 1100 K. The logarithmic axis highlights that the strongest enrichment effect occurs between pure methane and the first hydrogen-containing blend in this grid.

combustor, and it does not include flow, turbulence, recirculation, wall heat loss or geometry effects.

All 243 grid points ignited within the simulated time. Figure 6 shows that the combined temperature–equivalence-ratio dependence changes substantially with hydrogen addition. For methane–air the delay spans 14.28 ms at $T_0 = 1100$ K, $\phi = 0.6$ to 654.92 ms at $T_0 = 900$ K, $\phi = 1.4$. At 40 mol % hydrogen, the corresponding range is 1.26 ms at $T_0 = 1100$ K, $\phi = 1.4$ to 221.04 ms at $T_0 = 900$ K, $\phi = 0.6$. The maximum difference between the maximum- dT/dt delay criterion and the peak-OH criterion is 0.156%, so the two indicators are consistent for this diagnostic grid.

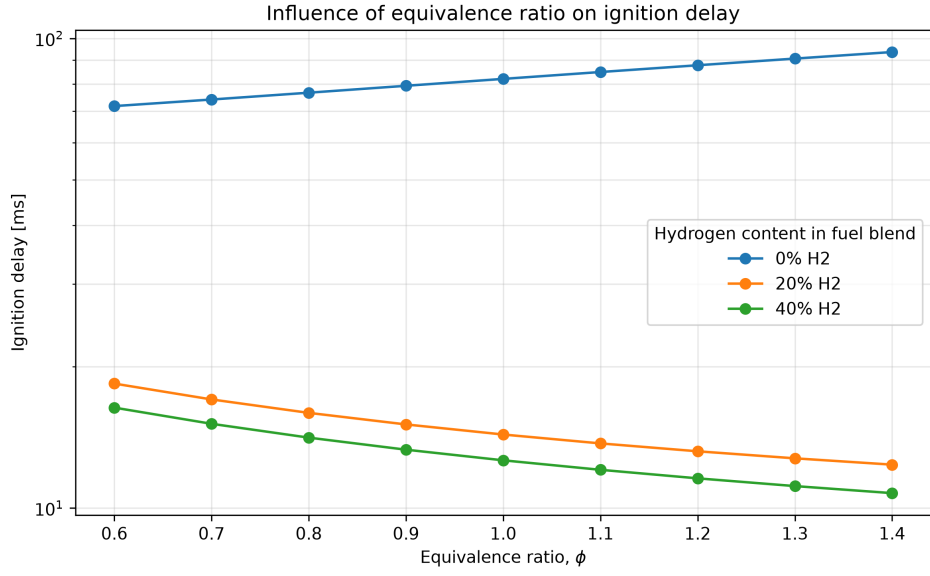
The ignition-delay reduction factor in Figure 7 is computed only where both cases ignite:

$$R_\tau = \frac{\tau_{\text{ign}, 0\% \text{ H}_2}}{\tau_{\text{ign}, 40\% \text{ H}_2}}. \quad (2)$$

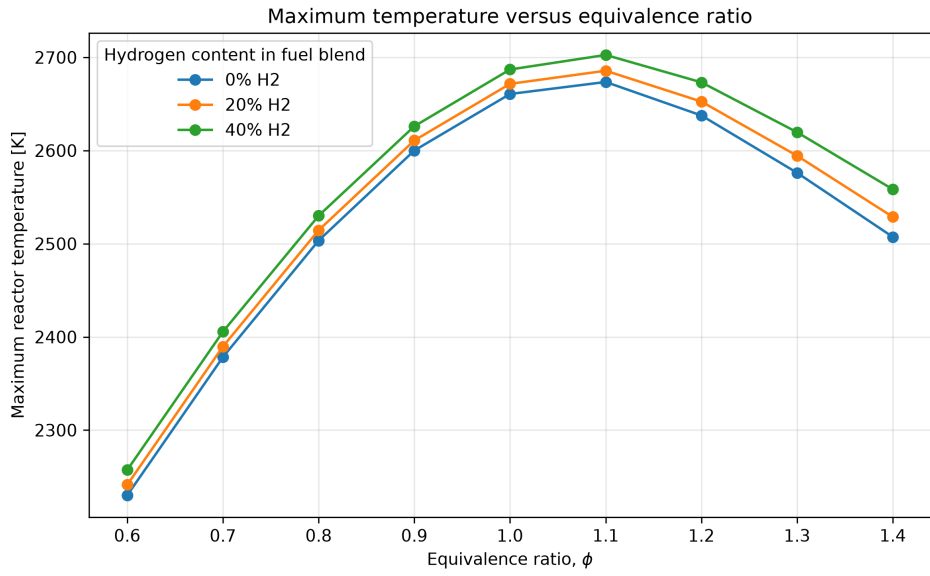
It ranges from 2.54 at $T_0 = 900$ K, $\phi = 0.6$ to 16.28 at $T_0 = 1100$ K, $\phi = 1.4$. Values greater than one mean that the calculated ignition delay is shorter after adding 40 mol % hydrogen to the fuel. This factor is a kinetic ignition-delay comparison only, not an engine efficiency, performance or emissions metric.

3.6 Constant-volume versus constant-pressure reactors

The reactor-model comparison isolates the effect of the constant-volume versus constant-pressure idealisation for the same mixtures and initial temperatures. For all tested cases, the constant-pressure ignition delay is longer by approximately 2.6% to 8.8%. The smallest difference is 2.60% at $T_0 = 900$ K and 0 mol % hydrogen (586.74 ms constant-volume versus 601.98 ms constant-pressure), while the largest is 8.84% at $T_0 = 1100$ K and 20 mol % hydrogen (2.05 ms versus 2.23 ms). At $T_0 = 1000$ K, the constant-pressure delays are 81.99 ms, 14.34 ms and 12.64 ms for 0, 20 and 40 mol % hydrogen, compared with constant-volume values of 79.74 ms, 13.41 ms and 11.86 ms. The constant-pressure reactor is therefore more representative of the adopted combustor-oriented idealisation, but it is not a complete gas-turbine model.



(a) Ignition delay.



(b) Maximum temperature.

Figure 5: Equivalence-ratio sweep at $T_0 = 1000$ K, $p = 10$ atm and hydrogen fractions of 0, 20 and 40 mol %.

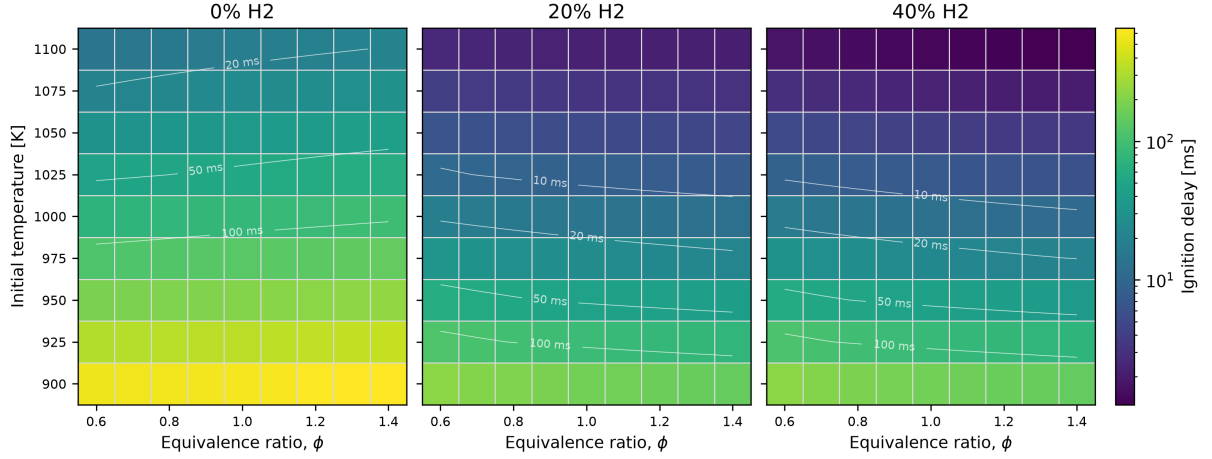


Figure 6: Two-dimensional constant-pressure ignition-delay map for CH_4/H_2 -air mixtures at $p = 10$ atm, $T_0 = 900$ K–1100 K, $\phi = 0.6$ –1.4 and fuel hydrogen fractions of 0, 20 and 40 mol %. Each panel uses the actual 9×9 grid, the colour scale is logarithmic and common to all panels, and the white labels mark constant ignition-delay isolines in milliseconds. This is an ideal homogeneous-reactor diagnostic, not an operational map of a real combustion chamber.

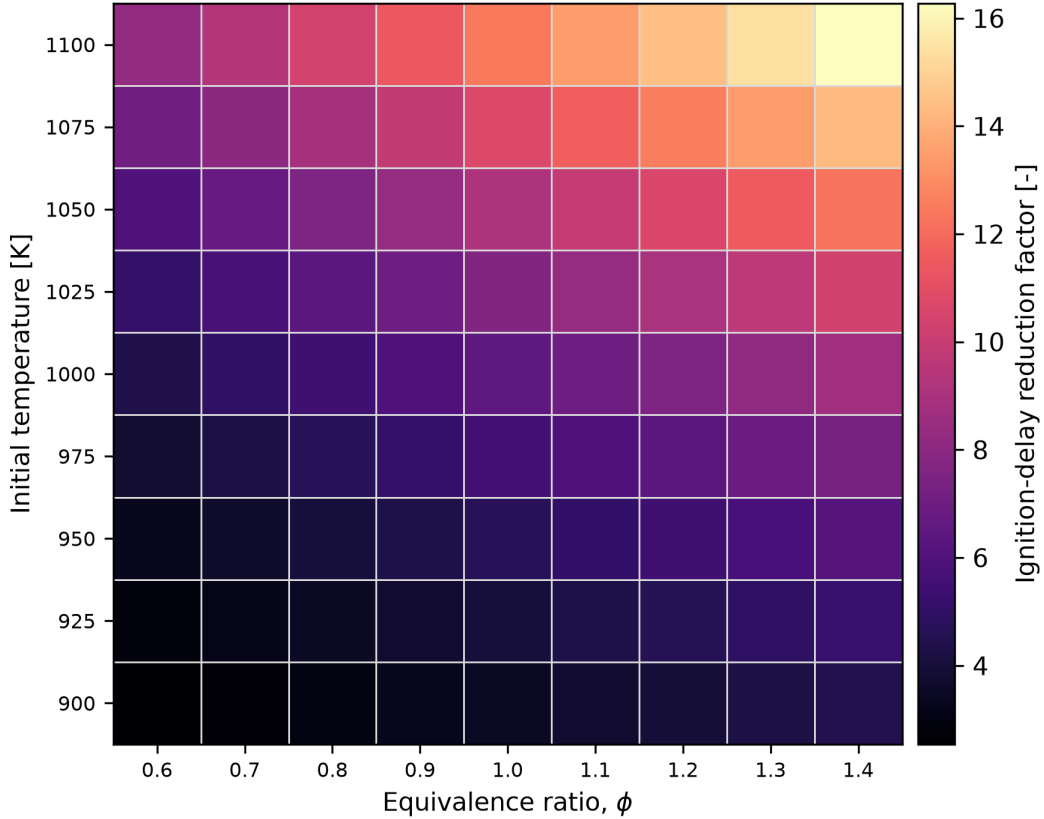


Figure 7: Ignition-delay reduction factor $R_\tau = \frac{\tau_{\text{ign}, 0\% \text{ H}_2}}{\tau_{\text{ign}, 40\% \text{ H}_2}}$ for the same T_0 - ϕ grid at $p = 10$ atm. Values above one indicate a shorter calculated ignition delay for the 40 mol % hydrogen blend; the quantity is not an engine-efficiency or combustor-performance metric.

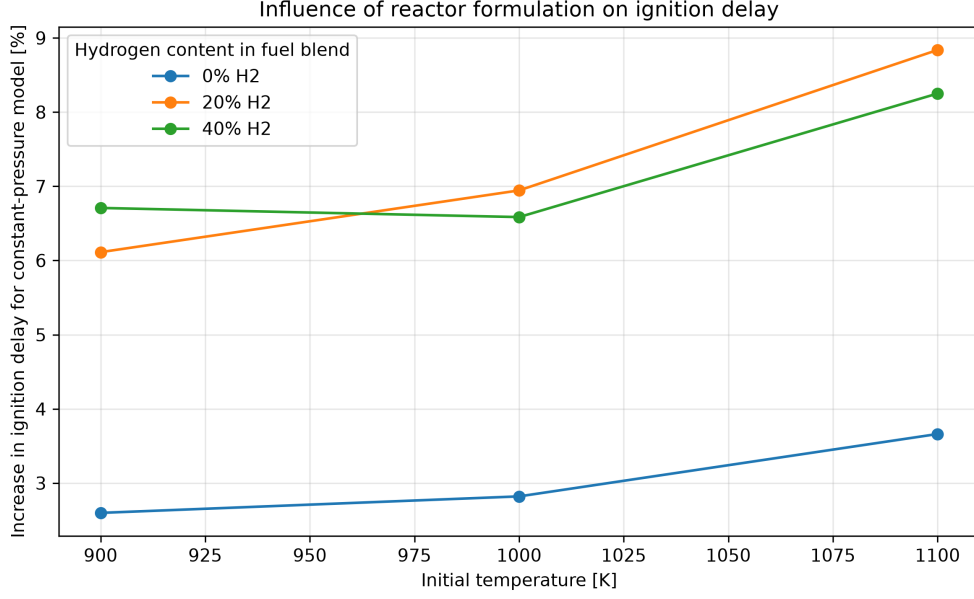
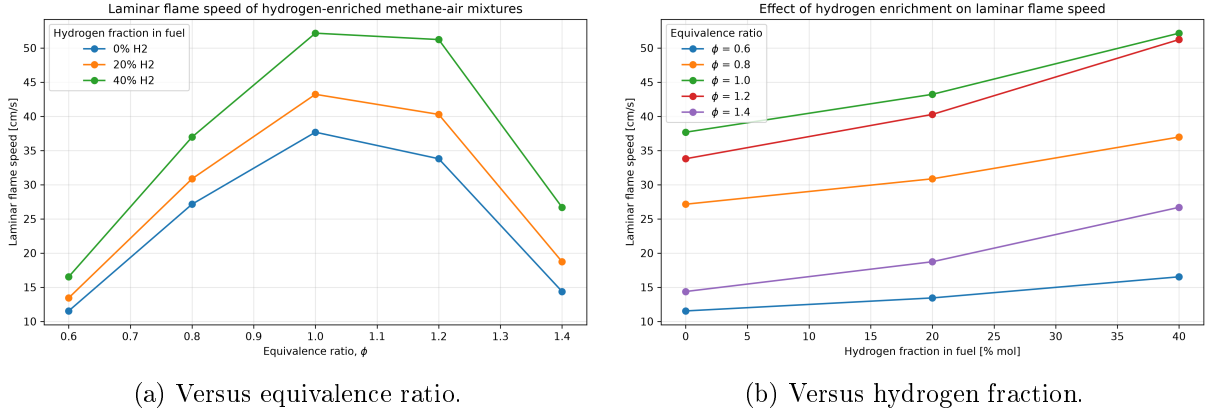


Figure 8: Ignition delay at $\phi = 1.0$ and $p = 10$ atm: constant-volume versus constant-pressure reactor for $T_0 = 900, 1000$ and 1100 K, and hydrogen fractions of 0, 20 and 40 mol %.



(a) Versus equivalence ratio.

(b) Versus hydrogen fraction.

Figure 9: Laminar flame speed of hydrogen-enriched methane-air flames.

3.7 Laminar flame speed

The computed laminar flame speed of stoichiometric methane-air at reference conditions is 37.68 cm s^{-1} . This value is reported as a result of the present **FreeFlame** model; no dedicated experimental validation was performed. Hydrogen enrichment increases the calculated stoichiometric flame speed to 52.16 cm s^{-1} at 40 mol % hydrogen in the fuel (Figure 9).

3.8 Grid-refinement check

The final flame-speed model uses multicomponent transport with Soret diffusion enabled, so the diagnostic grid-refinement check was repeated with the same transport settings rather than using earlier mixture-averaged calculations. Two representative stoichiometric cases were compared: methane-air and the 40 mol % hydrogen blend at 300 K and 1 atm. The current setting uses ratio = 3.0, slope = 0.06 and curve = 0.12; the tighter setting uses ratio = 2.5, slope = 0.03 and curve = 0.06.

For both representative cases, the current refinement settings differ from the tighter grid by less

Table 4: Diagnostic FreeFlame grid-refinement check using multicomponent transport and Soret diffusion.

Case	Grid	S_L [cm s ⁻¹]	Points	Difference vs. tight [%]
CH ₄	current	37.68	197	0.75
CH ₄	tight	37.40	359	0.00
40% H ₂	current	52.16	202	0.19
40% H ₂	tight	52.07	382	0.00

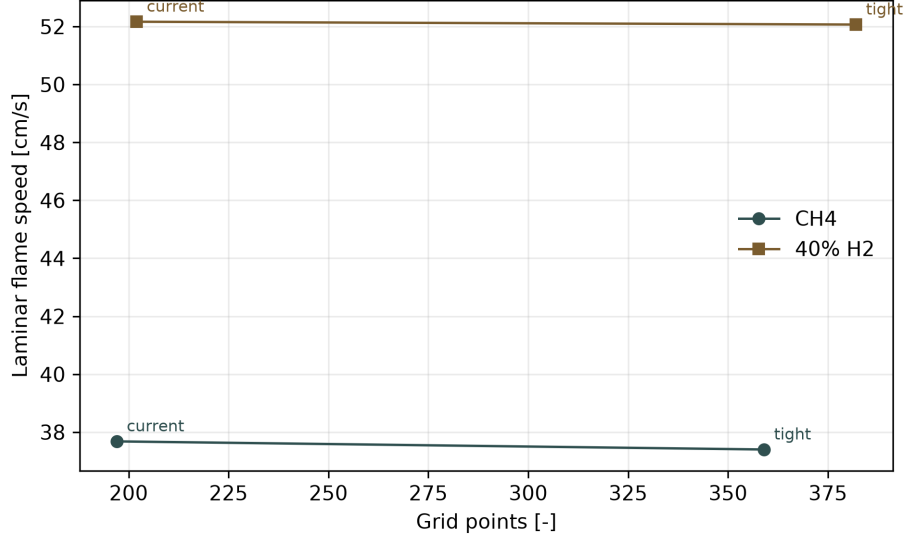


Figure 10: Diagnostic comparison of current and tight FreeFlame refinement settings. The check is a grid-refinement consistency check for two representative cases, not a validation against measurements.

than 1 %, indicating limited sensitivity of the reported flame speeds to further grid refinement.

3.9 Perfectly stirred reactor stability transition

For each tested fuel blend and equivalence ratio, the PSR calculations provide a stability-transition bracket between cold extinguished and hot stable steady-state solutions on the tested residence-time grid. This ideal PSR calculation does not represent actual engine behaviour, CFD, turbulence, combustor geometry, recirculation or wall heat loss.

Table 5: Discrete PSR stability-transition brackets on the tested residence-time grid.

ϕ	H ₂ in fuel [mol %]	Extinguished τ [ms]	Stable τ [ms]
0.7	0	0.005	0.010
0.7	20	0.002	0.005
0.7	40	0.002	0.005
1.0	0	0.002	0.005
1.0	20	0.002	0.005
1.0	40	0.002	0.005

These values are discrete brackets from the configured residence-time grid, not resolved transition times. Hydrogen enrichment moves the observed bracket for the lean $\phi = 0.7$ case from 0.005–

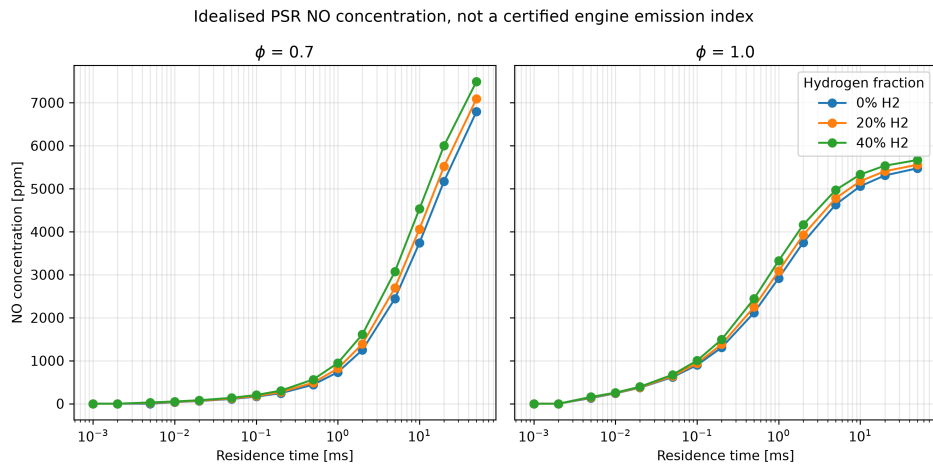
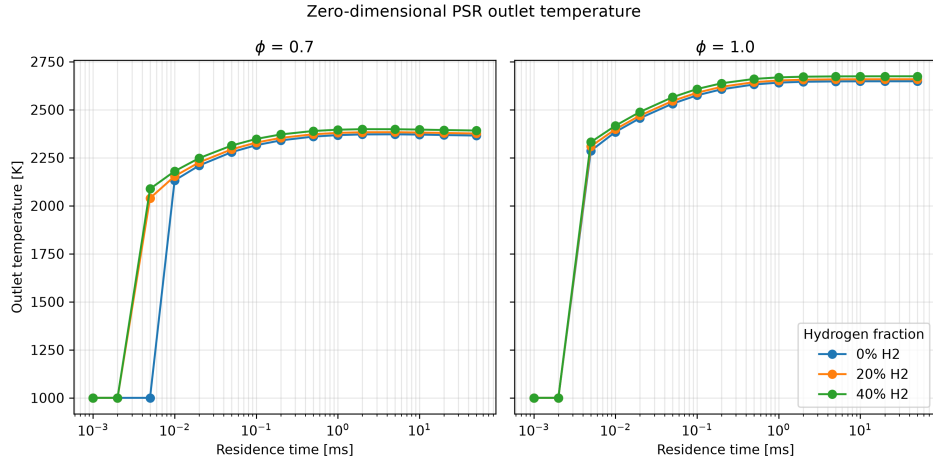


Figure 11: PSR behaviour versus residence time; the drop at short residence time marks a stability-transition bracket on the tested grid. Inlet conditions are $T_{\text{in}} = 1000 \text{ K}$, $p = 10 \text{ atm}$, with $\phi = 0.7$ and 1.0 .

0.010 ms for methane–air to 0.002–0.005 ms for the enriched cases; at $\phi = 1.0$, all three tested fuel blends share the same 0.002–0.005 ms bracket.

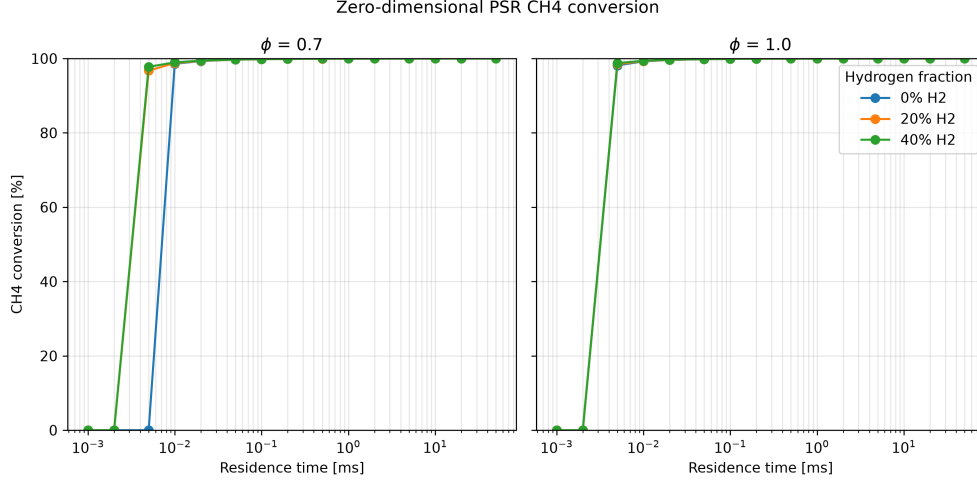


Figure 12: Ideal PSR CH_4 conversion versus residence time at $T_{\text{in}} = 1000 \text{ K}$, $p = 10 \text{ atm}$, $\phi = 0.7$ and 1.0 . Conversion is shown only as a finite-rate stirred-reactor trend; it does not include mixing, geometry, wall heat loss or turbine-combustor residence time distributions.

4 Model limitations and conclusions

4.1 Model limitations

GRI-Mech 3.0 was optimised and validated for natural-gas (methane-dominated) combustion [4]. For the hydrogen fractions considered here ($x_{\text{H}_2} \leq 0.4$), the mixture remains methane-dominated, so its use is defensible; however, for higher hydrogen content a dedicated hydrogen mechanism, such as that of Burke et al. [6] or Conaire et al. [7], would be more appropriate. The flame-speed calculations use multicomponent transport with Soret diffusion, but they still represent one-dimensional, freely propagating, adiabatic premixed flames. The PSR is zero-dimensional, adiabatic and perfectly stirred, with no representation of actual engine operation, CFD, turbulence, geometry, recirculation or wall heat loss. The equilibrium calculations also assume infinite-time chemical equilibrium and therefore cannot represent finite-rate effects in the nozzle/exhaust.

4.2 Conclusions

1. At $\phi = 1.0$, 300 K and 1 atm , increasing the fuel hydrogen fraction from 0% to 40% increased the calculated laminar flame speed from 37.68 cm s^{-1} to 52.16 cm s^{-1} .
2. At $T_0 = 1000 \text{ K}$, $p = 10 \text{ atm}$ and $\phi = 1.0$, the constant-pressure ignition delay decreased from 81.99 ms for methane-air to 12.64 ms for the 40% hydrogen blend.
3. For the tested PSR grid, cold extinguished and hot stable steady-state solutions were separated by a stability-transition bracket in residence time, rather than by a resolved exact transition point.
4. The present results are limited to detailed-chemistry idealisations: equilibrium reactors, homogeneous ignition calculations, one-dimensional freely propagating flames and a zero-dimensional adiabatic PSR.
5. In a simplified combustor-screening sense, hydrogen enrichment in the tested CH_4/H_2 blends tends to accelerate ignition and increase flame speed, so it should be evaluated together with stability margins, thermal loading and emissions constraints before any hardware-level interpretation.

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